

MCQs:-

1. A ✓
2. B ✓
3. C ✓
4. D ✓
5. A ✓
6. C ✓
7. C X
8. A ✓
9. B ✓
10. C ✓
11. A ✓
12. D ✓

Method :-

Q: 2(i)(or)
A: 2(i)(or)

A measuring cylinder is partly filled with water and initial water level is noted. The irregular solid is then lowered in so that it is fully submerged and the new (higher) water level is noted. Since the solid displaces its own volume of water. The volume of the solid = final reading - initial reading.

Initial water :- 24 ml

Final water : 63 ml

Volume : $63 \text{ ml} - 24 \text{ ml} = 39 \text{ ml} = 39 \text{ cm}^3$.

Q: 2(ii)

A: 2(ii)

(a) 0.000085 m

$8.5 \times 10^{-5} \text{ m}$

$85.0 \times 10^{-1} \times 10^{-5} \text{ m}$

$85 \times 10^{-6} \text{ m}$

$85 \mu\text{m}$

(b) $52,000,000 \text{ s} = 5.2 \times 10^7 \text{ s}$

Q. 2 (iii)
A. 2 (iii)



$$a = \frac{v_f - v_i}{t}$$

$$= \frac{26 - 6}{8}$$

$$= \frac{20}{8} \Rightarrow 2.5 \text{ m/s}^2$$

$$b) s = vt + \frac{1}{2}at^2$$

$$(6) \times 8 + \frac{1}{2}(2.5) \times (8)^2$$

$$48 + \frac{1}{2}(2.5)(64)$$

$$48 + \frac{1}{2}(160)$$

$$48 + \frac{80}{1}$$

$$48 + 80$$

$$128 \text{ m}$$

Q: 2(iv)
A: 2(iv)
Total path length travelled (distance is scalar, direction ignored)
distance: $500\text{m} + 200\text{m} = 700\text{m}$

Total displacement:
 $S = 500\text{m} - 200\text{m} = 300\text{m}$

Q: 2(v)
A: 2(v)
Second (Newton) law:-

The acceleration of an object is directly proportional to the magnitude of net force, in a same direction is inversely proportional to the mass of an object.

Equation:- $F = ma$

(a) $F = 42\text{N}$, $m = 7\text{Kg}$

$F = ma$

~~acceleration~~ $a = \frac{F}{m} = \frac{42}{7} \Rightarrow 6\text{m/s}^2$

(b) Same force, $m = 3.5\text{Kg}$

$$a = \frac{F}{m} \Rightarrow \frac{42}{3.5} = 12\text{m/s}^2$$

$$\begin{array}{c} \text{Q:2(vi)} \\ \text{A:2(vi)} \end{array}$$

$$(a) \Delta p = m(v_f - v_i)$$

$$\begin{array}{l} 0.16 \times (30 - 0) \\ 0.16(30) \Rightarrow 4.8 \text{ Kg} \cdot \text{m/s} \end{array}$$

$$(b) F = \frac{\Delta p}{\Delta t}$$

$$= \frac{4.8}{0.02} = 240 \text{ N}$$

$$\begin{array}{c} \text{Q:2(vii)} \\ \text{A:2(vii)} \end{array}$$

The clockwise moment is equal to the anticlockwise moment about the same Pivot :-

$$\text{Rabia's moment} = \text{Adeel's moment}$$

$$300 \text{ N} \times 0.8 \text{ m} = 400 \text{ N} \times d$$

$$240 \text{ N} \cdot \text{m} = 400 \text{ N} \times d$$

$$\frac{240}{400} = d$$

$$0.6 = d$$

$$0.6 = d$$

Q: 2(vii)(or)
A: 2(vii)(or)
The centre of gravity (COG) of a body is the point at which the entire weight of the body may be considered to act.

Method for an irregular lamina:-
Make a small hole near the edge of the lamina and suspend it freely from a pin, so it can swing.

Hang a plumb line (a string with a small weight) from the same pin, and mark the line it traces on the lamina.

Repeat this from at least 2 more different points on the lamina, each time marking the vertical line traced by the plumb line.

The point where all these lines intersect is the centre of gravity of the lamina.

Q: 2(iii)(or)

A: 2(iii)(or)

When 2 hands are rubbed together, the rough surfaces of the skin resist relative motion, this resistance is ~~called~~ friction. This rubbing does work against friction, this work is ~~disipated~~ dissipated as heat.

which is why the hands feel warmer.

Energy conversion :-

the mechanical energy used in moving the hands back and forth is converted into thermal energy.

$$Q: 2(n)$$

$$A: 2(n)$$

Using $P = \rho gh$

$$800 \times 10 \times 4 = 32,000 \Rightarrow 3.2 \times 10^4$$

$$Q: 2(n)$$

$$A: 2(n)$$

Pressure is defined as the force acting normally (perpendicularly) per unit area.

$P = \frac{F}{A}$. SI unit is pascal (Pa)

A

$$1 \text{ Pa} = 1 \text{ N/m}^2$$

force (weight of tractor) :-

$$F = mg = 2500 \times 10 = 25,000 \text{ N}$$

$$P = \frac{F}{A} = \frac{25000}{0.5} = \frac{50,000}{5.0 \times 10^4}$$

Equation

$$v = v_i + a t$$

$$s = v t + \frac{1}{2} a t^2$$

$$2 a s = v_f^2 - v_i^2$$

1

$$v_i = 0$$

h =

2

Q.3 (i) (or)
A. 2 (i) (or)

Equation

Symbols

$$v = v_i + at$$

$$s = vt + \frac{1}{2}at^2$$

$$2as = v_f^2 - v_i^2$$

v_i = initial velocity

v_f = final velocity

a = acceleration

t = time

s = distance covered.

$$v_i = 0, g = 10, t = 5s$$

$$h = 0(5) + \frac{1}{2}(10)(5)^2$$

$$0 + \frac{1}{2}(10)(25)$$

$$125$$

$$0 + \cancel{250}$$

$$= 125m$$

$$v_f = v_i + gt$$

$$= 0 + (10)(5)$$

$$50 \text{ m/s}$$

Q. 3(ii) (or)
A. 3(ii) (or)

Impulse is the product of net force acting on a body and the time for which it acts. Impulse = $F \Delta t$

Derivation from Second Newton's Law

$$F = ma$$

$$F = m \left(\frac{v_f - v_i}{t} \right)$$

$$F = \frac{mv_f - mv_i}{t}$$

$$F = \frac{\Delta p}{\Delta t}$$

$$F \Delta t = \Delta p$$

(b) $m = 1.2 \text{ kg}$, $v_i = 12 \text{ m/s}$, $v_f = 0 \Rightarrow \Delta t = 0.008$

(i) Impulse = $\Delta p = m(v_f - v_i)$

$$= 1.2(0 - 12)$$

$$= -14.4 \text{ kg.m/s}$$

magnitude of impulse = 14.4 N.s

$$F = \frac{\Delta p}{\Delta t}$$

$$= \frac{-14.4}{0.008}$$

$$= -1800 \text{ N}$$

Derivation of

Consider a
sectional
p. It's
mass is m.
 $F = mg = P$
base of
the bar
P.

(b) i)

ii) J

$$\frac{F}{A} = \frac{\Delta P}{\Delta h}$$

$$= \frac{14.4}{0.008}$$

$$= 1800 \text{ N}$$

$$\text{Derivation of } P = \rho gh$$

Consider a liquid column of cross-sectional area A and height h , density ρ . It's volume is $V = Ah$. So it's mass is $m = \rho Ah$ and it's weight is $F = mg = \rho Ahg$. This weight acts on the base of area A . So the pressure at the base is:-

$$P = \frac{F}{A}$$

$$\frac{\rho Ahg}{A} = \rho hg$$

$$\begin{aligned} \text{b) i) } 800 \times 10 \times 0.5 &= 4000 \text{ Pa} \\ \text{ii) } P(\text{bottom}) &= P(\text{oil}) + P(\text{water}) \cdot g \cdot h(\text{water}) \\ &= P(\text{bottom}) \end{aligned}$$

$$\frac{4000 + 1000(2.5)P(\text{bottom})}{4000 + 25000} = 2900 \text{ Pa}$$

Q: 3(iv)(or)

A: 2(iv)(or)

Renewable

Non-Renewable

① Naturally replenished over time & cannot run out (e.g. geothermal heat from the Earth's core & tidal energy from the Moon's pull) limited in amount, formed over millions of years & cannot be replaced once used (e.g. coal, natural gas, petroleum).

Example:-

biomass

Example:-

nuclear fuel

② PE lost per second = mgh
 $= 2000 \times 10 \times 15$
 $300,000 \text{ J}$

③ Power = $\frac{\text{Energy}}{\text{time}} = \frac{300,000}{1}$

$= 300,000 \text{ W} = 300 \text{ kW}$